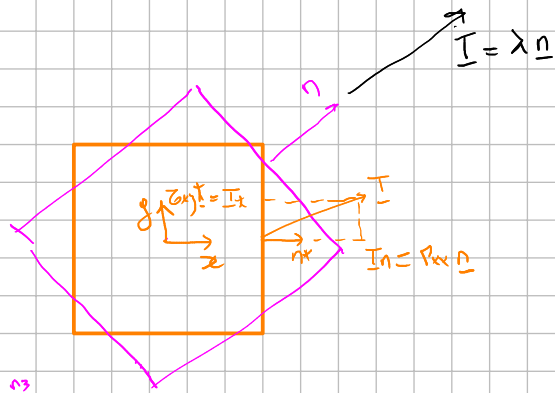


Quiz 4 - Egl

F. Czychy

$$\underline{I} = \underline{y} \cdot \underline{n} = \lambda \underline{n}$$

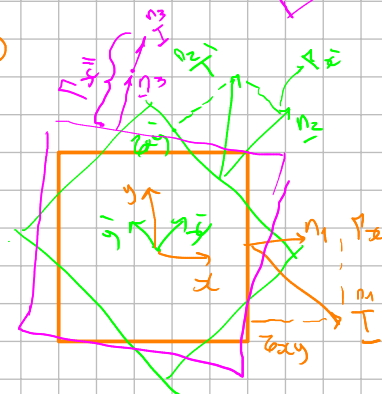
$$\underline{I} // \underline{n}$$



$\{x'x', y'y', z'z'\}$ new axes
 $\rightarrow$ orthogonal
 $\hookrightarrow z'x' = 0$

$$\underline{I}' = \underline{P} \underline{I} \underline{P}^T$$

$$\begin{bmatrix} x'x' & z'x' \\ z'x' & y'y' \end{bmatrix} = \underline{P} \begin{bmatrix} x'x' & z'x' \\ z'x' & y'y' \end{bmatrix} \underline{P}^T$$



$$(x, y) \xrightarrow{\underline{P}} (x', y')$$

$$\underline{I} = \underline{I}' \cdot \underline{n}'$$

$$\underline{P}^T \underline{I}' = \underline{I}' \underline{P}^T \underline{n}'$$

$$\underline{I}' = \underline{P} \underline{I} \underline{P}^T \underline{n}'$$

Rotación de 1 tensor de 2do orden de tensores para un problema 2D

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta, \quad (4.2-6)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta, \quad (4.2-7)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta. \quad (4.2-8)$$

From these equations, it follows that

Centro del círculo de Mohr
abscisa del centro del círculo de Mohr

$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y \quad (4.2-9)$$

Máximas tensiones normales:

$\sigma_{\max}$ y $\sigma_{\min}$

$\frac{\partial \sigma_{x'}}{\partial \theta} = 0 \wedge \frac{\partial \sigma_{y'}}{\partial \theta} = 0$

$$\frac{\partial \sigma_{x'}}{\partial \theta} = 2\tau_{x'y'}, \quad \frac{\partial \sigma_{y'}}{\partial \theta} = -2\tau_{x'y'}, \quad (4.2-10)$$

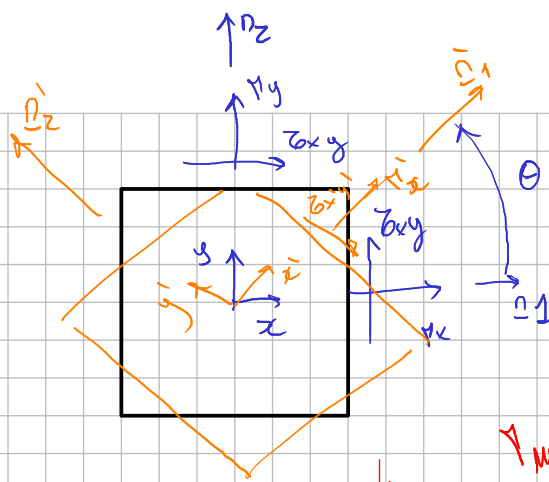
$$\tau_{x'y'} = 0 \text{ when } \tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}. \quad (4.2-11)$$

$$\sigma_{\max} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\min}$$

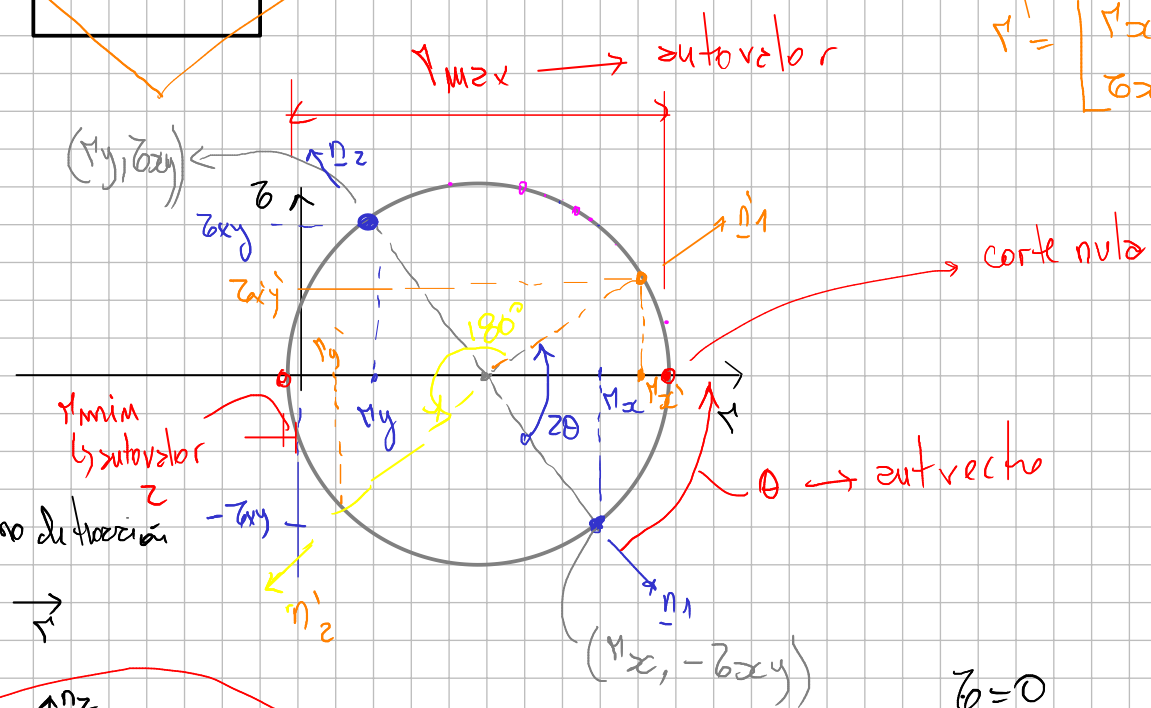
setting the derivative equal to zero, we can find its extreme value. It can be shown that this angle is $\pm 45^\circ$ from the principal directions given by Eq. (4.2-11) and that the maximum value of $\tau_{x'y'}$ is

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

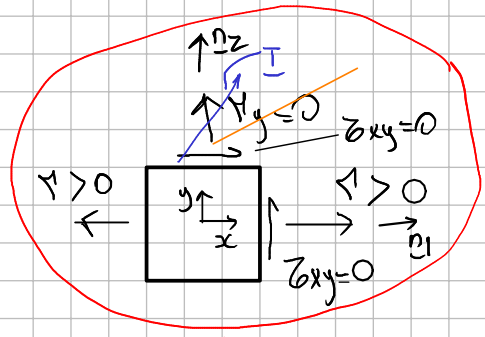
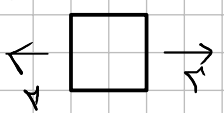


$$T = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

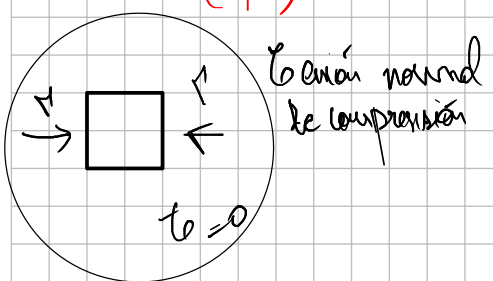
$$T' = \begin{bmatrix} \cos \theta' & -\sin \theta' \\ \sin \theta' & \cos \theta' \end{bmatrix}$$



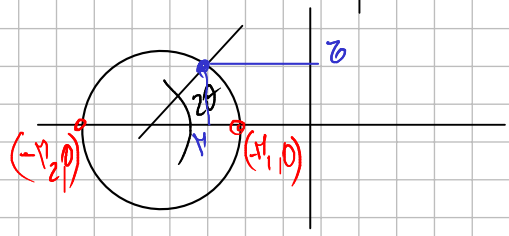
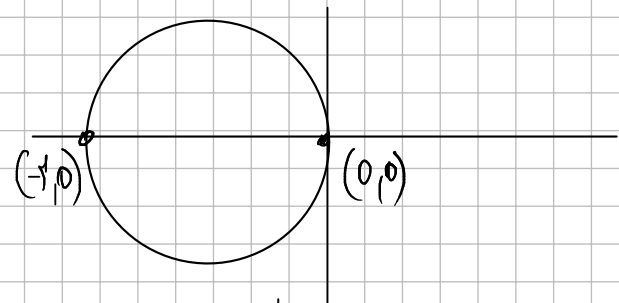
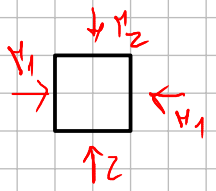
Sección normal de tracción



$(\sigma_1, 0)$
 $(\sigma_2, 0)$



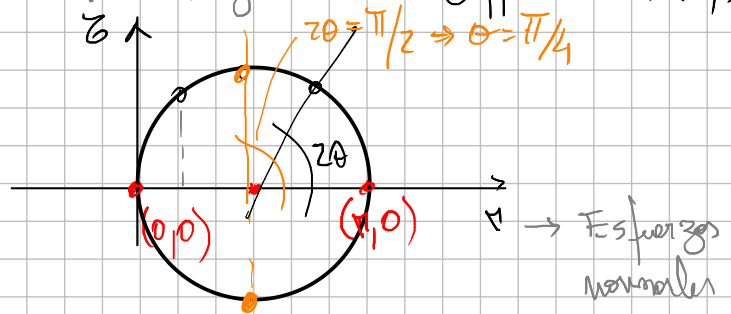
Sección normal de compresión



$\sigma = 0$

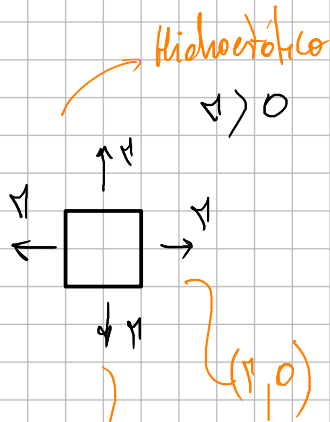
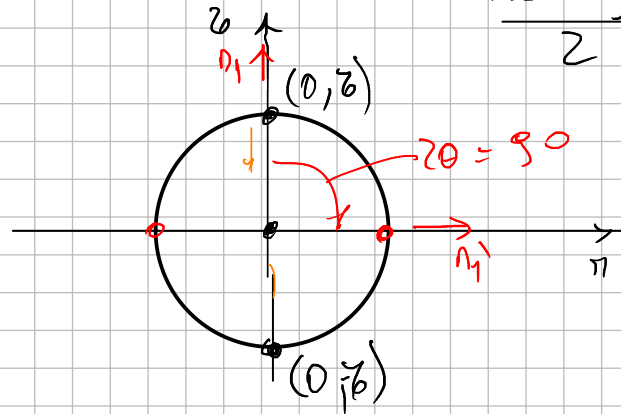
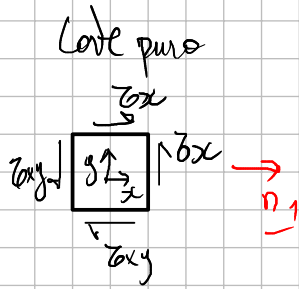
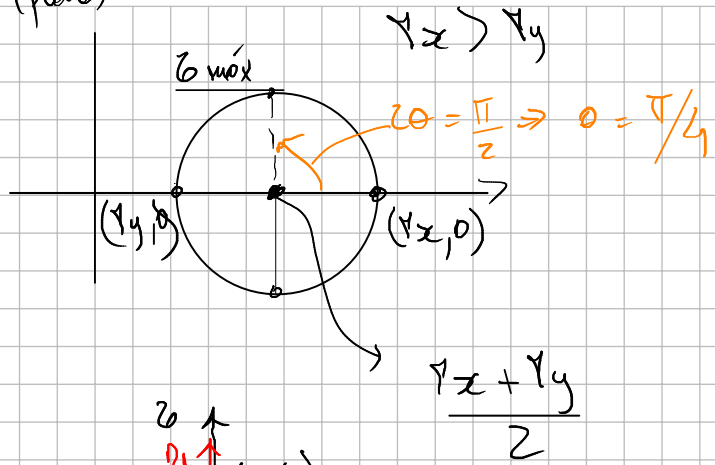
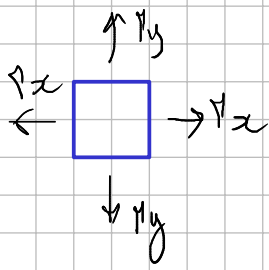
Esferas longitudinales

$\theta_{pp} = 0 \rightarrow (x, y) \rightarrow$ dir. ppales



Esferas normales

Tensión normal máxima (plano)



σ_x

$$\sigma = \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix}$$

